

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

Next item →

1. Suppose that a continuous-time system has $A = -0.1$, $B = 0.1$, $C = 3$, $D = 0$, $B_w = 1$, $S_w = 0.5$, and $S_v = 0.01$. If the sample period is $\Delta t = 0.1$, what is A_d ? You may use the Coursera Labs Jupyter notebook to compute the answer to this question. Enter your result with two digits to the right of the decimal point.

1 / 1 point

0.9900498

✔ Correct
Yes. This is correct.

2. Suppose that a continuous-time system has $A = -0.1$, $B = 0.1$, $C = 3$, $D = 0$, $B_w = 1$, $S_w = 0.5$, and $S_v = 0.01$. If the sample period is $\Delta t = 0.1$, what is $\Sigma_{\tilde{w}}$? You may use the Coursera Labs Jupyter notebook to compute the answer to this question. Enter your result with four digits to the right of the decimal point.

1 / 1 point

0.04950332

✔ Correct
Yes. This is correct.

3. Suppose that a continuous-time system has $A = -0.1$, $B = 0.1$, $C = 3$, $D = 0$, $B_w = 1$, $S_w = 0.5$, and $S_v = 0.01$. If the sample period is $\Delta t = 0.1$, what is $\Sigma_{\tilde{v}}$? You may use the Coursera Labs Jupyter notebook to compute the answer to this question. Enter your result with two digits to the right of the decimal point.

1 / 1 point

0.1

✔ Correct
Yes. This is correct.